36 zeros. Given the potential trillions of devices that would be online with the Internet of Things, this seems like a safe upper limit which can manage all the addresses.

Another element of the network architecture which supports the Internet of Things is the Domain Name System or DNS. The system is used to name things connected to any network. Its most apparent function is to translate human readable domain names to numeric IP address for devices. It also works to index the entire network directory of names which follow the IP suite. The stability of DNS over current Internet systems has proven to be high but in the anticipated era of the Internet of Things there is still uncertainty. The question of whether DNS infrastructures currently in place will be able to sustain the load of trillions of devices is untested. The issues of performance, latency and reaction speeds is still being figured out to not only make DNS scalable beyond its current limitations but to foresee any potential gaps in its security given the flood of new devices that it is sure to encounter.

Building IoT: Proliferation, connection and intelligence

One of the most amazing aspects of the Internet of Things is sheer variety of function and form. Potentially any device could be a part of this vast network - from something as simple as a toothbrush to something as complex as a car. However, the means to make a truly useful system between them can only be designed when a clear intent and infrastructure is in place. Over the last few years, as the concept of the Internet of Things, has been tested in smaller settings such as factories and consumer systems like smart homes, a series of steps have been suggested by experts which are essential for a fully functional Internet of Things system.

The first step of establishing and building the Internet of Things is to further the proliferation of sensory devices. Since the system is powered by data obtained by "things" it becomes important for the world to have as many of these "things" as possible. The problem is that there is no limit to what device can support data sensors which raises the question of intent. Already companies are experimenting with the potential use of these devices in the world, e.g. Coca Cola purchased 16 million MAC addresses for future use on their "things". While at present only a fraction of these 16 million devices are being used by the company, mostly in digital vending machines, there is a scope of them expanding to unknown areas as well. Similarly other devices such as shoes, apparels, televisions, appliances and many other